Chapter 6 — Edge Colouring

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1 Edge Colouring

Definition 1.1 (Edge Colouring)**.** *An **edge colouring** of a graph $G = (V, E)$ is an assignment of colours to the edges of G such that no two edges incident to the same vertex share the same colour:*

$$c : E \to \{1, 2, \ldots, k\}, \quad \text{with } c(e_1) \neq c(e_2) \text{ whenever } e_1, e_2 \text{ share a vertex.}$$

*A **proper edge colouring** satisfies this condition. The minimum number of colours required is called the **edge chromatic number**, denoted $\chi'(G)$.*

Remark 1.1. *Edge colouring is dual to vertex colouring via the **line graph**: given G, its line graph $L(G)$ has vertices corresponding to edges of G, and two vertices of $L(G)$ are adjacent if the corresponding edges in G share a vertex. Then a proper edge colouring of G is equivalent to a proper vertex colouring of $L(G)$.*

1.1 Examples of Edge Colouring

Example 1.1 (Triangle K_3)**.** *All three edges are incident with each other. Therefore, $\chi'(K_3) = 3$.*

Example 1.2 (Square C_4)**.** *Edges can alternate between two colours, so $\chi'(C_4) = 2$.*

Example 1.3 (Star Graph $K_{1,n}$)**.** *Maximum degree $\Delta = n$, so $\chi'(K_{1,n}) = n$ since each edge meets at the central vertex.*

2 Edge Chromatic Number

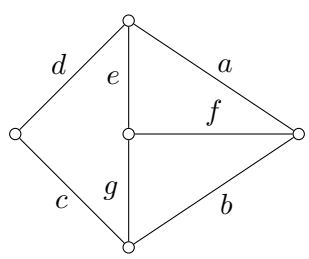


Figure 6.1

Definition 2.1 (Edge Chromatic Number)**.** *The **edge chromatic number** (or **chromatic index**) $\chi'(G)$ is the smallest number of colours required to properly colour the edges of G.*

2.1 Basic Properties and Bounds

- **Lower bound:**

  $$\chi'(G) \geq \Delta(G)$$

  because the edges incident to a vertex of maximum degree must all have distinct colours.

- **Upper bound (Vizing's Theorem):**

  $$\chi'(G) \leq \Delta(G) + 1$$

  This gives rise to the classification:

  - **Class 1:** $\chi'(G) = \Delta(G)$
  - **Class 2:** $\chi'(G) = \Delta(G) + 1$

3 Vizing's Theorem

Theorem 3.1 (Vizing, 1964)**.** *For any simple graph G:*

$$\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1$$

Remark 3.1. *This theorem ensures that every simple graph can be edge-coloured using at most one more than the maximum degree of the graph.*

3.1 Proof Sketch

- Colour edges sequentially.
- If no colour is available for a new edge, construct an **alternating path** between two colours.
- Swap colours along this path to free a colour.
- Repeat until all edges are coloured using at most $\Delta + 1$ colours.

3.2 Examples

Example 3.1 (Bipartite Graph $K_{3,3}$)**.** $\Delta = 3$, $\chi'(K_{3,3}) = 3$ *(Class 1)*

Example 3.2 (Complete Graph K_5)**.** $\Delta = 4$, $\chi'(K_5) = 5$ *(Class 2)*

4 The Timetabling Problem

Definition 4.1 (Timetabling Problem)**.** *Given a set of exams and students, schedule exams so that no student has overlapping exams.*

4.1 Graph Theoretical Formulation

- Represent exams as vertices.
- Connect vertices with an edge if a student attends both exams.
- Each time slot corresponds to a colour in edge colouring.
- Construct the line graph $L(G)$ of the conflict graph G: vertices of $L(G)$ correspond to edges of G, adjacent if edges in G share a vertex.
- Then $\chi(L(G)) = \chi'(G)$, reducing timetabling to edge colouring.

5 König's Line Colouring Theorem

Theorem 5.1 (König, 1916)**.** *For any bipartite graph G:*

$$\chi'(G) = \Delta(G)$$

Remark 5.1. *All bipartite graphs are Class 1. The theorem guarantees a proper edge colouring using exactly $\Delta(G)$ colours.*

5.1 Proof Sketch

- Use induction on the number of edges.
- Apply Hall's Marriage Theorem to find matchings for each colour.
- Colour edges sequentially, ensuring no conflict at vertices.

5.2 Example

$K_{3,3}$ is bipartite with $\Delta = 3$, so by König's theorem, $\chi'(K_{3,3}) = 3$.

6 Summary Table

Concept	**Definition / Statement**	**Notes**
Edge Colouring	Assignment of colours to edges so adjacent edges differ	Basis for chromatic index
Edge Chromatic Number (χ')	Minimum colours needed	$\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1$
Vizing's Theorem	$\chi'(G) \leq \Delta(G) + 1$	Class 1 vs Class 2
Timetabling Problem	Scheduling exams / tasks	Uses line graphs and edge colouring
König's Line Colouring Theorem	$\chi'(G) = \Delta(G)$ for bipartite graphs	Always Class 1

7 Applications

- Scheduling problems (exams, tournaments, shifts)
- Register allocation in compilers
- Traffic flow management
- Frequency assignment in networks

8 The Timetabling Problem

8.1 Problem Statement

The **timetabling problem** models the scheduling of exams, classes, or tasks in such a way that no conflict occurs. Formally, consider:

- T teachers or instructors,
- C classes or exams,
- A **teaching requirement matrix** $P = [p_{ij}]$, where $p_{ij} = 1$ if teacher i teaches class j, and 0 otherwise.

We represent this problem as a **bipartite graph** $G = (T \cup C, E)$, where:

- Vertices correspond to teachers T and classes C.
- An edge $e = (t, c)$ exists if teacher t teaches class c (i.e., $p_{tc} = 1$).

A **timetable** corresponds to an assignment of edges to **periods** such that:

- No teacher is scheduled to teach more than one class in a period.
- No class is taught by more than one teacher in a period.

Equivalently, each period corresponds to a **matching** in the bipartite graph G, since matchings are sets of edges with no two edges incident to the same vertex.

8.2 Balancing Matchings

Lemma 8.1. *Let M and N be disjoint matchings of G with $|M| > |N|$. Then there exist disjoint matchings M' and N' of G such that:*

$$|M'| = |M| - 1, \quad |N'| = |N| + 1, \quad \text{and } M' \cup N' = M \cup N.$$

Proof Sketch:

- Consider the subgraph $H = G[M \cup N]$, the union of edges in M and N.
- Each component of H is either:
  - An even cycle, with edges alternating between M and N, or
  - A path with edges alternating between M and N.
- Since $|M| > |N|$, some path component P must start and end with edges of M.
- Let $P = v_0 e_1 v_1 e_2 \cdots e_{2n+1} v_{2n+1}$. Then define:

$$M' = (M \setminus \{e_1, e_3, \ldots, e_{2n+1}\}) \cup \{e_2, e_4, \ldots, e_{2n}\},$$

$$N' = (N \setminus \{e_2, e_4, \ldots, e_{2n}\}) \cup \{e_1, e_3, \ldots, e_{2n+1}\}.$$

- This switches edges along the path, reducing $|M|$ by 1 and increasing $|N|$ by 1 while preserving disjointness.

8.3 Theorem: Existence of p Disjoint Matchings

Theorem 8.2. *Let G be a bipartite graph with maximum degree A. If $p \geq A$, then there exist p disjoint matchings $M_1, M_2, \ldots, M_p$ of G such that:*

$$\bigcup_{i=1}^{p} M_i = E(G), \tag{1}$$

$$\text{and any two matchings differ in size by at most one: } ||M_i| - |M_j|| \leq 1. \tag{2}$$

Interpretation: Each matching corresponds to a period in the timetable, so Theorem 6.3 guarantees:

- Every edge (teacher-class assignment) is scheduled in some period.
- Workload is balanced across periods.

8.4 Example: Teachers and Classes

Suppose there are 4 teachers and 5 classes with the teaching requirement matrix P, and that one possible 4-period timetable is shown in Figure 6.4.

- The edge decomposition of G assigns edges to matchings:
  - Normal edges $\to$ period 1,
  - Broken edges $\to$ period 2,
  - Wavy edges $\to$ period 3,
  - Heavy edges $\to$ period 4.
- Period 1 requires 4 rooms (4 edges in the matching). Total edges $|E| = 11$.
- By Theorem 6.3, a 4-period timetable can be arranged such that each period has either $\lceil 11/4 \rceil = 3$ or $\lfloor 11/4 \rfloor = 2$ classes.

8.5 Refining the Timetable

- Let M_1 be the normal matching (period 1), M_4 the heavy matching (period 4), with $|M_1| = 4$ and $|M_4| = 2$.
- Consider $G[M_1 \cup M_4]$, which consists of two path components of length 3.
- Using Lemma 6.3, we can **swap edges along the paths**:
  - Reduce M_1 by 1 edge and increase M_4 by 1 edge.
  - Balance the matchings so each period has 2 or 3 classes.
- This reduces the maximum number of rooms needed at any period from 4 to 3.

Remark 8.1. *This procedure generalizes to any bipartite timetabling problem: given $p \geq \Delta(G)$ periods, the edges of the bipartite graph can be decomposed into p disjoint matchings with roughly equal sizes, minimizing peak resource usage (e.g., rooms).*

$$P = \begin{pmatrix} 2 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

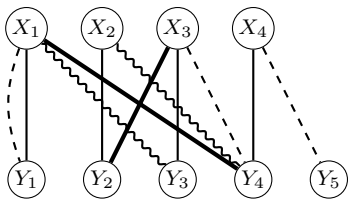
(a)

	Period			
	1	2	3	4
X_1	Y_1	Y_1	Y_3	Y_4
X_2	Y_2	-	Y_4	-
X_3	Y_3	Y_4	-	Y_2
X_4	Y_4	Y_5	-	-

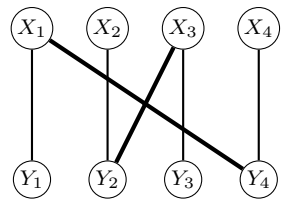
(b)

Figure 6.4

$$\lfloor \epsilon/p \rfloor \leq |M_i| \leq \lceil \epsilon/p \rceil$$

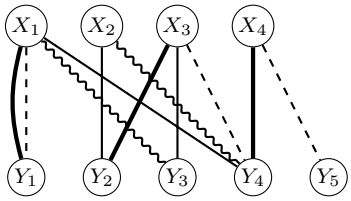


(a)



(b)

Figure 6.5



(a)

	Period			
	1	2	3	4
X_1	Y_4	Y_1	Y_3	Y_1
X_2	Y_2	-	Y_4	-
X_3	Y_3	Y_4	-	Y_2
X_4	-	Y_5	-	Y_4

(b)

Figure 6.6 Period order: (1)normal, (2)dash, (3)wiggle, (4)thick